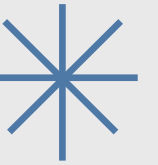


2025 DOMESTIC FLIGHT ANALYSIS

Ava McCann



BUSINESS PROBLEM



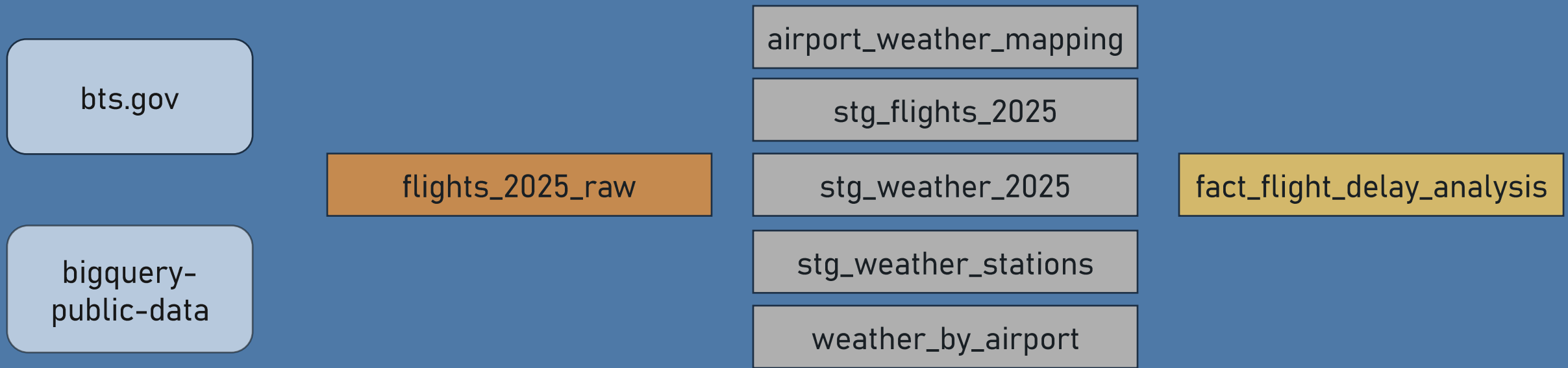
Airline delays and cancellations create operational inefficiencies, customer dissatisfaction and increase costs. Understanding when and where disruptions occur can help identify high-risk areas and improve airline reliability.

North Star Metric:

On-Time Rate

Key Questions:

- Do certain airlines and airports operate more efficiently?
- What factors are top contributors to delays? Are delays and flight volume correlated?
- Are certain airports and routes higher risk?



METHODS

- Extracted 2025 domestic flight data from Bureau of Transportation Statistics.
- Extracted 2025 weather, weather station and airport data from BigQuery public data
- Implemented Medallion Architecture to clean, organize and structure data.
 - Mapped weather stations to the nearest airport to connect weather and flight data.
 - Integrated all flight, weather and airport data into OBT as a final output
- Final dashboard visualization created in Tableau

2025 Domestic Flight Data

Airports

Airlines

All
Airports

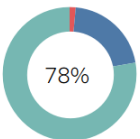
TOTAL FLIGHTS
7,001,618



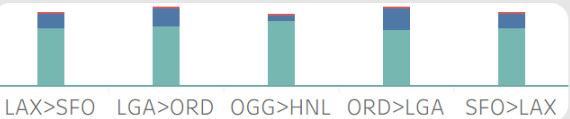
AVG DELAY
13.56



ON-TIME RATE



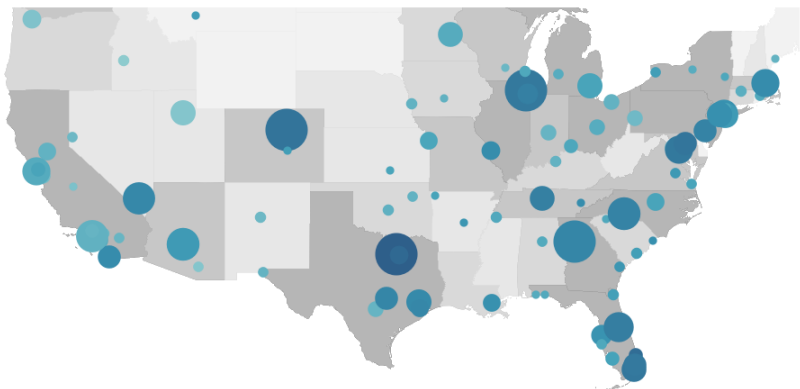
TOTAL ROUTES
6,938



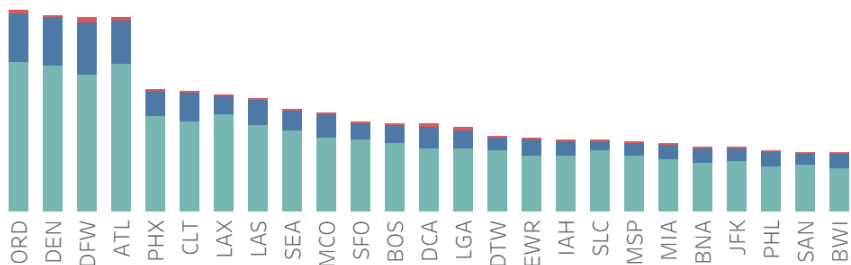
Delay Rate

0.1078 0.2776 All

Month of Fl Date

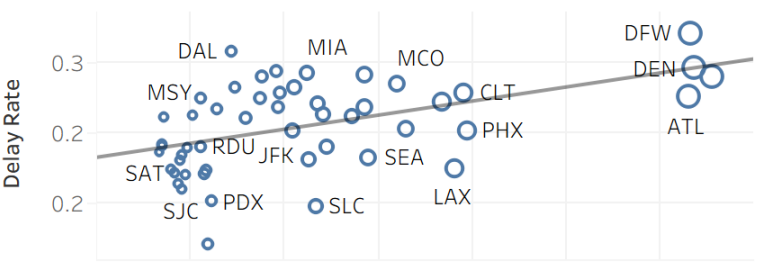
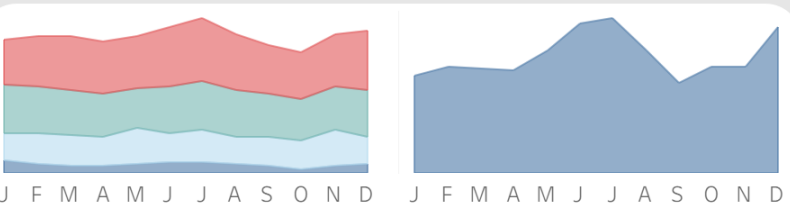
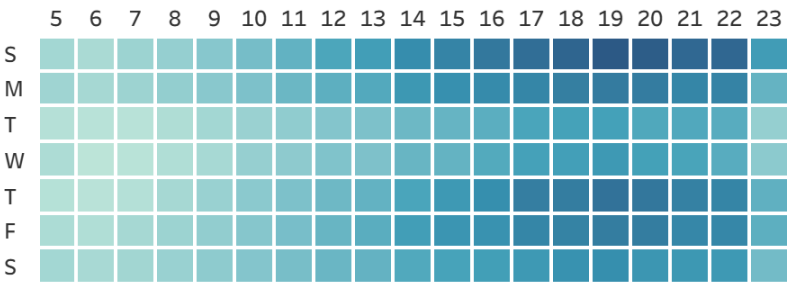


© 2026 Mapbox © OpenStreetMap



Average Delay by Weekday & Hour

Average Delay



INSIGHTS

Delays are highly seasonal

- Delay rates & times raise during summer months and winter holidays, loosely correlated with total flight volume.

Delays are correlated with busier airports

- Airport volume and delay rate show a positive relationship.

Delays compound throughout the day

- Flight volume stays steady while delays get longer, especially from 4-10pm. Low flight volume from 12am-5am allows for the delay time to be reset.

2025 Domestic Flight Data

Airports

Airlines

Delta

TOTAL FLIGHTS

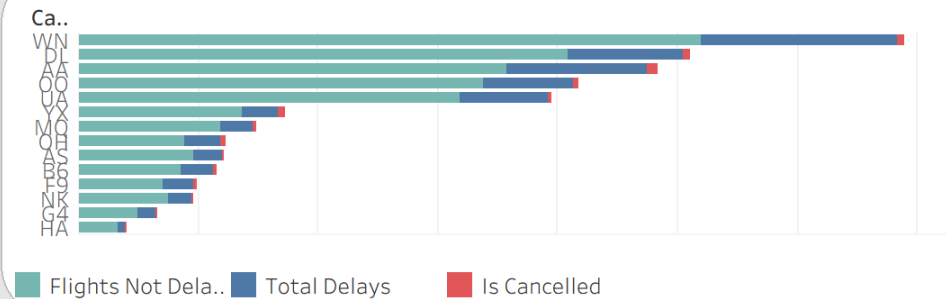
1,026,332

TOTAL ROUTES

1,167

ON-TIME RATE

80%



Filters

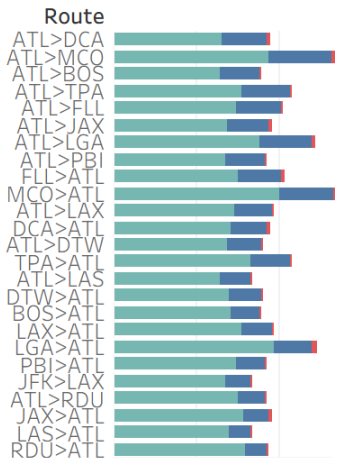
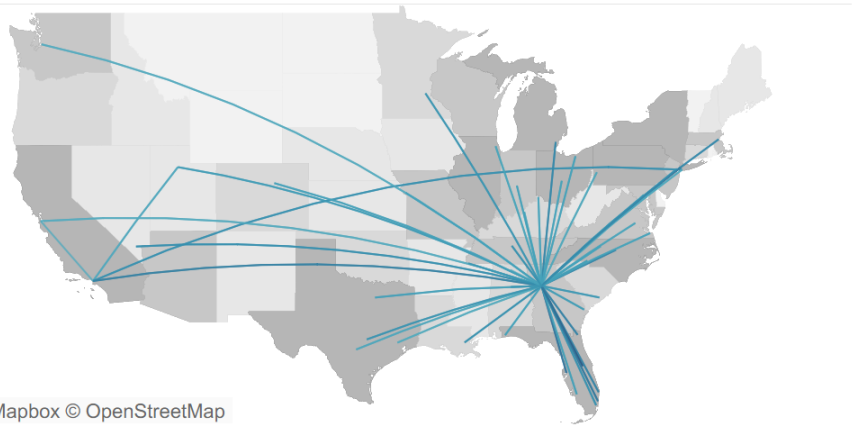
Carrier Code

DL

Month

All

Total Flights 0 5,375



Jan 12 Feb 9 Mar 9 Apr 6 May 4 Jun 1 Jun 29 Jul 27 Aug 24 Sep 21 Oct 19 Nov 16 Dec 14

Week [2025]

INSIGHTS

Top 5 airports dominate flight volume

Airline flight volume is not a strong predictor of delays

- Top airlines have more routes and more flights, but their reliability varies and does not correlate with their volume.

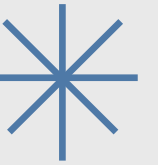
Route popularity is not a strong predictor of delays

- Top routes have widely varying delay rates with little correlation to their volume for the airline.

Delay risk for routes is related to airports

- Delays correlate the strongest with specific airports, especially ones with higher volumes and other high-risk airports.

CONCLUSION



Final Insights

- Flight delays and cancellations exhibit seasonal and hourly patterns.
- Airport congestion and certain bottleneck airports cause higher delay rates.
- Airline size and route volume alone are not reliable predictors of on-time performance, and it is more strongly correlated with airport volume.

Further Questions

To what extent are airline delay rates driven by the airports & routes they serve versus airline operational inefficiency?

Next Steps

- Develop predictive models for delay rates & cancellations
- Incorporate additional & current years of data through an automated data pipeline